

Bluestock Mutual Fund Analytics

Final Project Report

Prepared as a technical analytics submission covering data ingestion, cleaning, database design, exploratory analysis, performance analytics, risk analytics and dashboard development.

Important scope note: The available project data contained six mutual-fund NAV series. Investor transaction, AUM, portfolio-holdings and external benchmark datasets were not available in the working project files, so unsupported values were not fabricated.

1. Executive Summary

This project develops an end-to-end mutual-fund analytics workflow using Python, Pandas, NumPy, SQLite and Power BI. The workflow covers raw NAV data preparation, validation, database loading, exploratory analysis, return and risk measurement, advanced risk metrics and dashboard presentation.

The working dataset contains six scheme-level NAV histories: HDFC Top 100 Direct, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip and Kotak Bluechip. The cleaned NAV history contains 20,025 observations.

The project also includes performance metrics, a fund scorecard, VaR/CVaR output, Alpha/Beta analysis, a rolling 90-day Sharpe chart and a simple recommender script.

- Use reproducible Python scripts instead of hard-coded manual calculations.
- Keep unsupported investor and industry-wide figures separate from scheme-level NAV analysis.

2. Project Objectives

The primary objective is to create a reproducible analytical pipeline for mutual-fund NAV data and expose the resulting insights through a Power BI dashboard.

The project was designed around ETL, relational data modeling, analytical SQL, EDA, performance measurement, risk measurement and reporting.

- Clean and validate NAV history.
- Load analytical data into SQLite.
- Calculate returns and risk metrics.
- Build a Power BI dashboard.
- Produce advanced analytics and submission documentation.

3. Data Sources and Scope

The raw-data directory contains scheme CSV files and an NAV history source. Six scheme datasets were available for the current implementation.

The project also used MFAPI as the intended source for live NAV retrieval. Scheme identifiers used in the project include 125497, 119551, 120503, 118632, 119092 and 120841.

The scope is therefore a six-scheme analytical implementation rather than a fabricated 40-scheme or 1,908-scheme market-wide study.

- HDFC Top 100 Direct — 125497
- SBI Bluechip — 119551
- ICICI Bluechip — 120503
- Nippon Large Cap — 118632
- Axis Bluechip — 119092
- Kotak Bluechip — 120841

4. Folder and Repository Structure

The project is organized into raw data, processed data, database files, notebooks, SQL scripts, Python scripts, dashboard assets and reports.

Git is used for version control. The repository contains separate commits for data cleaning, dashboard work, documentation and advanced analytics.

- data/raw
- data/processed
- data/db
- notebooks
- scripts
- sql
- dashboard
- reports

5. ETL and Data Ingestion

The ingestion stage reads NAV CSV files into Pandas DataFrames and standardizes the date and NAV fields. The pipeline sorts observations by scheme and date and produces a consolidated NAV history.

The cleaned NAV history contains 20,025 rows with three core columns. The processing stage is designed to make downstream calculations reproducible.

- Use Pandas for ingestion and transformation.
- Convert dates to datetime.
- Sort observations by scheme and date.
- Remove invalid or duplicate observations where applicable.

6. Data Quality and Validation

Validation was performed against the available fund master information. The working validation covered the six scheme codes used in the analysis.

The available validation result showed all six working scheme codes were present, with zero missing codes in the NAV history validation.

- Fund-master validation: 6 working scheme codes.
- Missing working scheme codes: 0.
- Cleaned NAV history: 20,025 rows.
- The project does not claim completeness for datasets that were not supplied.

7. Database Design

A SQLite database was created to provide a structured analytical store. The star-schema design separates descriptive dimensions from measurable facts.

The database contains dim_fund, dim_date, fact_nav, fact_transactions, fact_performance and fact_aum tables. The available NAV implementation populated the fund, date and NAV components; unavailable transaction, performance and AUM source datasets were not invented.

- dim_fund — scheme attributes.
- dim_date — date dimension.
- fact_nav — daily NAV observations.
- fact_transactions and fact_aum remain dependent on source datasets.

8. Exploratory Data Analysis

The EDA notebook analyzes NAV trends, daily returns, annualized volatility, cumulative growth and return correlations across the available schemes.

The analysis focuses on observable scheme-level NAV behavior rather than presenting market-wide AUM, SIP, folio or demographic statistics that require additional datasets.

- NAV trend analysis.
- Daily return distribution.
- Annualized volatility.
- Correlation analysis.
- Cumulative growth comparison.
- Best and worst daily observations.

9. Performance Analytics

Performance analysis derives daily returns from sequential NAV observations. Annualized return and volatility are calculated from the daily series.

The project includes Sharpe ratio, maximum drawdown and other performance fields. Advanced analytics also exports a consolidated `advanced_metrics.csv` and `fund_scorecard.csv`.

- Daily return = percentage change in NAV.
- Annualization uses a 252-trading-day convention where appropriate.
- Maximum drawdown measures decline from a running NAV peak.

10. Risk Analytics

Risk analysis includes volatility, maximum drawdown, historical Value at Risk and Conditional Value at Risk. VaR at the 95% level is based on the lower-tail historical return quantile; CVaR is the mean of observations at or below that threshold.

A rolling 90-day Sharpe series was also generated to show how risk-adjusted performance changes through time.

- Historical VaR 95%.
- Historical CVaR 95%.
- Rolling 90-day Sharpe.
- Maximum drawdown.

11. Alpha and Beta Analysis

The project exports `alpha_beta.csv`. The current implementation uses an available NAV return series as the comparison proxy because a dedicated Nifty 100 benchmark dataset was not present in the working files.

This limitation is important: benchmark-relative conclusions should only be made after a validated benchmark series is supplied.

- Output: `data/processed/alpha_beta.csv`.
- Benchmark limitation documented rather than hidden.
- A dedicated Nifty benchmark should be added for a full benchmark-relative implementation.

12. Fund Scorecard

A fund scorecard is included to provide a compact comparison of the available schemes. The current working scorecard is based on the available advanced risk-adjusted metrics.

The scorecard should be interpreted only within the six-scheme dataset and should not be treated as a market-wide ranking.

- Output: data/processed/fund_scorecard.csv.
- Scope: six available schemes.
- Interpretation should consider data coverage and benchmark limitations.

13. Advanced Analytics and Recommender

The advanced analytics stage produces `advanced_metrics.csv`, `var_cvar_report.csv`, `alpha_beta.csv` and `rolling_sharpe_chart.png`. A `recommender.py` script groups the available schemes by a simple risk-grade construction and returns up to three schemes for a selected risk appetite.

The recommender is a demonstration of rule-based analytics, not personalized financial advice.

- Risk appetite input: Low, Medium or High.
- Selection uses the available Sharpe metric.
- Only the six available schemes are eligible.

14. Power BI Dashboard

The Power BI dashboard contains four analytical pages: Executive Overview, Performance Analysis, Risk Analysis and Fund Comparison.

The dashboard uses scheme/date filtering and presents NAV trends, return, volatility, maximum drawdown, Sharpe and Beta metrics for the available schemes.

- Interactive filtering by scheme and date.
- Performance comparison.
- Risk comparison.
- Fund-level metric table.

15. Limitations and Data Gaps

Several requirements in the original specification depend on datasets that were not available in the working project directory. These include investor transactions, full AUM history, portfolio holdings, demographics, state-level transactions, folio counts and a dedicated Nifty benchmark series.

The current submission therefore avoids fabricating values such as market-wide AUM, SIP inflows, folio counts or the number of schemes in the entire industry.

- Six schemes rather than 40.
- No fabricated investor or AUM data.
- No fabricated portfolio sector weights.
- Benchmark-relative analysis should be expanded when benchmark data is available.

16. Reproducibility and Version Control

The project uses Git for version control and keeps analytical outputs in predictable project folders. The repository includes commits covering data cleaning, dashboard work, documentation and advanced analytics.

The v1.0 tag was created for the project state before the final documentation additions. Subsequent documentation changes should be tagged separately if strict release reproducibility is required.

- Git repository maintained on main.
- Advanced analytics committed.
- README documentation added.
- Generated report and presentation should be committed after creation.

17. Conclusion

The completed implementation demonstrates an end-to-end mutual-fund analytics workflow using real scheme-level NAV observations. It combines ETL, data validation, relational storage, statistical analysis, risk measurement and dashboard visualization.

The main technical strength of the submission is reproducibility and explicit handling of missing source data. Future expansion should focus on adding the unavailable investor, AUM, holdings and benchmark datasets so that the remaining market-level requirements can be calculated from source evidence.

Appendix: Key Deliverables

Component	Current output
Clean NAV history	data/processed/nav_history_cleaned.csv
Performance metrics	data/processed/performance_metrics.csv
Advanced metrics	data/processed/advanced_metrics.csv
Fund scorecard	data/processed/fund_scorecard.csv
VaR/CVaR	data/processed/var_cvar_report.csv
Alpha/Beta	data/processed/alpha_beta.csv
Rolling Sharpe	data/processed/rolling_sharpe_chart.png
Recommender	scripts/recommender.py
Power BI	dashboard/bluestock_mf.pbix